

07 Sep 2026 21:54:17 ET | 19 pages

US Semiconductors and Semiconductor Equipment

2Q26 Earnings Recap: Summer Sell-off; What's Next?

CITI'S TAKE

After an 18% summer sell-off in SOX where the group ran into high expectations and a modest deceleration in earnings momentum, we expect a rebound going into the Citi Global TMT conference where we expect positive semis commentary led by AMD, MRVL, LITE, ALAB, and semi caps (see [preview and key themes](#)). Fundamentally, data center demand remains strong (34% of semis TAM), auto/industrial (21%) continues to recover, while demand from PCs and handsets (42%) remains weak due to memory cost inflation and supply constraints. We lift our CPU TAM to \$237B by 2030 and see 2028 WFE bull case approaching \$300B post earnings.

Average EPS revisions slower for semis than semi caps: — C26/27 consensus EPS estimates for 2Q26 reports were up 5%/10% on average for semis and 10%/11% for semi caps. Estimate revision momentum slowed down for semis from 1Q 15%/12% and maintained for semi caps. Networking/analog/semi caps earnings momentum remained strong, followed by compute, and then memory (Figure 3).

Raising CPU TAM to \$237B by 2030 — We update our CPU model post earnings and Hot Chips conference. We believe the CPU TAM expands from \$29B in 2025 to \$237B in 2030, or a 52% CAGR. This forecast is above the forecasted TAMs from AMD, Arm, and NVIDIA or \$100B/\$220B/\$200B respectively. Post Hot Chips, we are more constructive on the innovation occurring in the server CPU space. We still expect AMD to be the key beneficiary of the CPU renaissance and Intel as a secondary beneficiary.

2028 WFE Bull Case Approaching \$300B — We update our top-down WFE analysis for 2026/2027/2028 based on [Citi's updated hyperscaler capex model](#) of 106%/56%/33% growth in CY26/27/28. We now see base case WFE of \$158Bn/\$209Bn/\$257Bn, with bull case WFE reaching \$163Bn/\$224Bn/\$286Bn in 2026/27/28 (including \$3Bn/\$10Bn/\$10Bn Terafab spending), up from \$147Bn/\$200Bn/\$250Bn prior and implies 37%/27% growth in both 2027/2028 in the bull case.

Atif Malik^{AC}

+1-415-951-1892

atif.malik@citi.com

Elizabeth Sun, CFA

+1-212-816-3308

elizabeth.sun@citi.com

James Bowlin

+1-415-951-1790

james.bowlin@citi.com

Kelsey Chia, CFA

kelsey.chia@citi.com

Papa Sylla

papa.sylla@citi.com

^{AC}See Appendix A-1 for Analyst Certification, Important Disclosures and Research Analyst Affiliations.

2Q26 Earnings Recap

Figure 1. Consensus EPS Estimates Following 2Q26 Earnings

	2026 Consensus EPS			2027 Consensus EPS		
	Pre-earnings	Post-earnings	Delta	Pre-earnings	Post-earnings	Delta
Semiconductors			5%			10%
ADI	\$12.86	\$13.53	5%	\$15.38	\$17.24	12%
ALAB	\$3.07	\$3.99	30%	\$4.84	\$6.49	34%
AMD	\$7.51	\$7.63	2%	\$13.85	\$15.77	14%
AVGO	\$12.94	\$12.90	0%	\$20.66	\$21.25	3%
CBRS	-\$0.90	-\$0.50	-44%	\$0.96	\$1.30	35%
GFS	\$1.90	\$1.94	2%	\$2.49	\$2.53	1%
INTC	\$1.11	\$1.54	39%	\$1.64	\$2.11	29%
IFX	\$2.30	\$2.32	1%	\$3.40	\$3.51	3%
MCHP	\$2.80	\$3.15	12%	\$3.92	\$4.33	11%
MPWR	\$24.17	\$27.42	13%	\$30.60	\$35.92	17%
MRVL	\$3.97	\$4.08	3%	\$6.12	\$6.52	7%
NXPI	\$14.85	\$15.12	2%	\$17.94	\$18.24	2%
NVDA	\$8.66	\$8.89	3%	\$12.52	\$15.19	21%
ON	\$3.09	\$3.20	3%	\$4.30	\$4.50	5%
QCOM	\$10.82	\$10.41	-4%	\$11.57	\$10.83	-6%
SLAB	\$2.50	\$2.91	16%	\$4.19	\$4.51	8%
SNDK	\$137.07	\$142.65	4%	\$226.67	\$238.29	5%
STM	\$1.32	\$1.36	3%	\$2.66	\$2.68	1%
SWKS	\$5.03	\$5.02	0%	\$5.37	\$5.17	-4%
TXN	\$7.82	\$8.53	9%	\$9.15	\$10.32	13%
Semi Cap			10%			11%
AEIS	\$9.42	\$11.24	19%	\$12.29	\$15.19	24%
AMAT	\$13.19	\$13.79	5%	\$18.21	\$19.63	8%
ASML	\$37.16	\$43.69	18%	\$51.60	\$60.14	17%
ENTG	\$3.66	\$3.92	7%	\$4.71	\$5.06	7%
FORM	\$2.46	\$3.05	24%	\$3.07	\$3.75	22%
KLAC	\$4.46	\$4.61	3%	\$5.81	\$6.07	5%
LRCX	\$6.97	\$7.65	10%	\$9.21	\$10.54	14%
MKSI	\$11.78	\$13.13	11%	\$15.45	\$17.50	13%
NVMI	\$10.48	\$10.67	2%	\$12.85	\$13.66	6%
TER	\$7.60	\$9.07	19%	\$10.46	\$11.41	9%
VECO	\$1.63	\$1.50	-8%	\$3.34	\$3.17	-5%

© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.

Source: Citi Research, FactSet

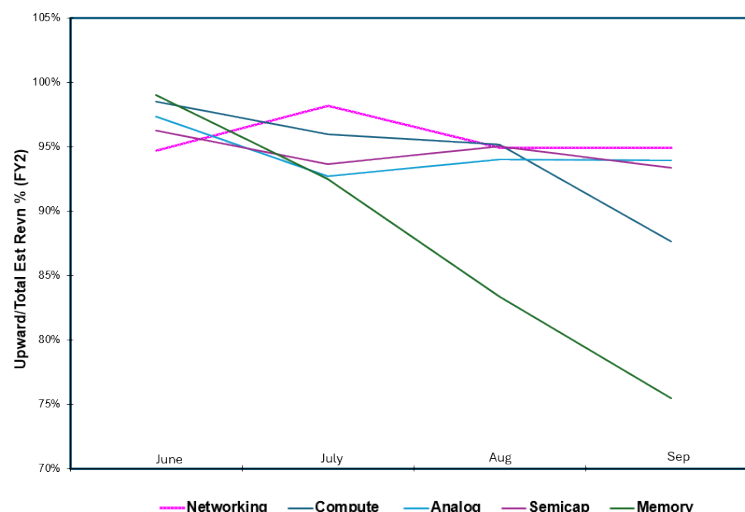
Figure 2. Consensus Sales Estimates Following 1Q26 Earnings

	2026 Consensus EPS			2027 Consensus EPS		
	Pre-earnings	Post-earnings	Delta	Pre-earnings	Post-earnings	Delta
Semiconductors			3%			7%
ADI	\$15,149	\$15,683	4%	\$17,310	\$18,847	9%
ALAB	\$1,562	\$1,909	22%	\$2,313	\$3,079	33%
AMD	\$50,196	\$50,995	2%	\$79,638	\$88,784	11%
AVGO	\$117,503	\$117,186	0%	\$185,268	\$191,646	3%
CBRS	\$863	\$887	3%	\$2,961	\$2,954	0%
GFS	\$7,244	\$7,307	1%	\$8,042	\$8,216	2%
INTC	\$58,801	\$63,213	8%	\$66,153	\$72,947	10%
IFX	\$19,499	\$19,903	2%	\$23,015	\$23,794	3%
MCHP	\$5,841	\$5,982	2%	\$6,944	\$7,174	3%
MPWR	\$3,713	\$4,149	12%	\$4,561	\$5,339	17%
MRVL	\$11,278	\$11,708	4%	\$16,413	\$17,681	8%
NXPI	\$14,136	\$14,289	1%	\$15,680	\$15,944	2%
NVDA	\$379,836	\$393,087	3%	\$556,682	\$670,059	20%
ON	\$6,481	\$6,551	1%	\$7,208	\$7,395	3%
QCOM	\$43,051	\$43,491	1%	\$46,046	\$46,694	1%
SLAB	\$945	\$926	-2%	\$1,087	\$1,081	-1%
SNDK	\$34,454	\$34,866	1%	\$52,733	\$53,951	2%
STM	\$14,351	\$14,405	0%	\$16,823	\$17,130	2%
SWKS	\$3,946	\$3,973	1%	\$4,098	\$4,104	0%
TXN	\$21,161	\$21,957	4%	\$23,623	\$25,012	6%
Semi Cap			6%			8%
AEIS	\$2,249	\$2,425	8%	\$2,699	\$3,009	11%
AMAT	\$35,227	\$36,418	3%	\$45,221	\$48,417	7%
ASML	\$45,664	\$49,786	9%	\$57,533	\$63,563	10%
ENTG	\$3,456	\$3,575	3%	\$3,886	\$4,039	4%
FORM	\$966	\$1,033	7%	\$1,113	\$1,197	8%
KLAC	\$15,459	\$15,876	3%	\$18,969	\$19,912	5%
LRCX	\$27,218	\$29,043	7%	\$33,964	\$37,971	12%
MKSI	\$4,817	\$5,081	5%	\$5,603	\$6,113	9%
NVMI	\$1,065	\$1,078	1%	\$1,296	\$1,360	5%
TER	\$4,550	\$5,139	13%	\$5,632	\$6,190	10%
VECO	\$779	\$800	3%	\$1,063	\$1,106	4%

© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.

Source: Citi Research, FactSet

Figure 3. Semis End Market Estimate Revisions



© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.

Source: Citi Research, FactSet

Raising CPU TAM to \$237B by 2030

CPU Background:

A CPU, or a central processing unit, is a type of processor that functions as the “brains” of a system and is commonly found in servers, PCs, smartphones, and various IoT devices. The server CPU market has historically been one of the most competitive and profitable applications for CPUs. For example, Intel’s Data Center segment typically had operating margins north of 40% for most of the 2010’s. Server CPUs come in a variety of flavors, including different core counts, ISA’s (instruction set architecture), channel counts, and various other factors.

Prior to the explosion of agentic AI, a CPU was primarily limited to two main applications: 1.) traditional workflows and 2.) acting as the head node for AI applications. In a head node, the CPU merely does the “management” by sending the user request to the GPU and sending the output back to the user. In this context, the GPU is doing heavy lifting such as running matrix multiplications to train LLMs or doing the inferencing when you query a chatbot. The third new application for CPUs is agentic AI, and this new application has seen explosive growth within the past year as AI has moved from chatbot inferencing to agentic autonomous workflows like Claude Code. At Citi, we define agentic CPUs as those that are handling the orchestration, reasoning loops, data processing, and security components of autonomous AI agents. In Agentic AI, CPUs have become the new bottleneck as CPU-to-GPU ratios move from 1:8 in model training, to 1:4 in inferencing, to 1:1 or greater in agentic AI. We believe the ratio will increasingly skew towards CPUs over time as agentic AI scales.

We believe AMD will be the primary beneficiary of the CPU renaissance due to their performance leadership, greater core counts, X86 ISA, broader SKU offerings, and support for both multithreading and single threading. In this framework, we compare the future offerings for Nvidia (Vera), AMD (Venice) and Intel (Diamond Rapids). AMD’s Venice CPU will offer core counts between 8 and 256 cores,

enabling AMD to support a variety of workloads while maintaining absolute performance leadership and supporting a variety of workloads. We believe Intel will have success as well, but we believe their Diamond Rapids CPU will trail AMD's Venice as it doesn't support multi-threading, has fewer threads, will be released ~2 quarters later, and expect an overall performance gap versus Venice. Intel's next generation CPU Coral Rapids will see a return to multi-threading, but this product likely won't be released until 2028. While we believe NVDA will be successful in reasoning server CPUs, Vera will feature a proprietary technology called spatial multi-threading, Vera will have fewer cores (Vera has 88 cores, Venice has up to 256 cores, Diamond Rapids has up to 256 cores), and Vera will utilize the Arm ISA. One of the drawbacks of the Arm ISA is that it faces software incompatibility with legacy systems and requires porting to run models smoothly, however this drawback is getting diminished over time. Given Nvidia is fabless and has a set allocation of wafers from its foundry partners, we believe Nvidia will prioritize its GPU shipments over individual CPU shipments given their entrenched position in the GPU market.

We update our CPU model post earnings and Hot Chips. We believe the CPU TAM expands from \$29B in 2025 to \$237B (previously \$137B) in 2030, or a 52% CAGR. We expect general purpose CPUs to grow at a 21% CAGR to \$52B in 2030, AI head nodes to grow at a 50% CAGR to \$63B in 2030, and agentic CPUs to grow at a 230% CAGR to \$123B in 2030. We model general purpose CPUs to represent 22% of the TAM in 2030, AI head nodes to represent 26% of the TAM in 2030, and agentic CPUs to represent 52% of the TAM by 2030. For reference, Arm forecasted a TAM of \$100B by 2030, AMD forecasted a TAM of \$220B by 2030, and Nvidia forecasted a TAM of \$200B.

Figure 4. Citi CPU TAM Model (\$M)

TAM Estimates (\$M)	C25	C26E	C27E	C28E	C29E	C30E
General Purpose CPU	\$20,544	\$27,310	\$33,706	\$37,929	\$47,522	\$52,226
% of Total	71%	53%	34%	29%	27%	22%
YoY		33%	23%	13%	25%	10%
AI Headnodes	\$8,185	\$12,522	\$27,577	\$34,395	\$47,199	\$62,646
% of Total	28%	24%	28%	26%	27%	26%
YoY		53%	120%	25%	37%	33%
Agentic CPU Applications	\$311	\$11,437	\$36,938	\$59,812	\$82,683	\$122,545
% of Total	1%	22%	38%	45%	47%	52%
YoY		3578%	223%	62%	38%	48%
Total	\$29,039	\$51,269	\$98,220	\$132,136	\$177,404	\$237,417
YoY		76%	35%	24%	30%	18%

© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.

Source: Citi Research, Mercury Research

With these updates, we now model total server CPU units to grow from ~30 million units in 2025 to 76 million units in 2030, or 21% CAGR (previously 17%). We believe Intel units could grow at 14% CAGR, AMD units could grow at 31% CAGR (previously 25%), and Arm units could grow at 26% CAGR (previously 25%).

Figure 5. Citi Server CPU Unit Model

Unit Model (M)	C25	C26E	C27E	C28E	C29E	C30E
CPU Units (M)	29,814	38,331	48,833	54,348	64,771	76,087
YoY	23%	29%	27%	11%	19%	17%
Intel	18,547	20,836	23,173	25,443	31,673	35,016
YoY	9%	12%	11%	10%	24%	11%
AMD	7,240	10,776	17,735	20,871	23,287	28,450
YoY	32%	49%	65%	18%	12%	22%
ARM/Others	4,027	6,719	7,924	8,034	9,812	12,621
YoY	127%	67%	18%	1%	22%	29%

© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.

Source: Citi Research, Mercury Research

We adjust our 2030 share estimates with our updated model. By 2030, we expect Intel to have 46% share (unchanged), AMD to have 37% share (previously 35%), and Arm/Others to have 17% share (previously 20%). We are more optimistic about AMD's near-term share opportunity with its leading TCO and supply allocation, as well as confidence in Intel ramping supply and improving yields. Most of our Arm shipment forecasts are coming from Nvidia Arm-based CPUs. We continue to expect AMD to be the primary beneficiary of the CPU renaissance, and Intel to be a secondary beneficiary given its manufacturing capacity and improving long-term roadmap.

Figure 6. Citi Server CPU Market Share Forecast

Unit Market Share (%)	C25	C26E	C27E	C28E	C29E	C30E
Intel	62%	54%	47%	47%	49%	46%
YoY	-780 bps	-785 bps	-690 bps	-64 bps	208 bps	-288 bps
AMD	24%	28%	36%	38%	36%	37%
YoY	163 bps	383 bps	820 bps	208 bps	-245 bps	144 bps
ARM/Others	14%	18%	16%	15%	15%	17%
YoY	617 bps	402 bps	-130 bps	-144 bps	37 bps	144 bps

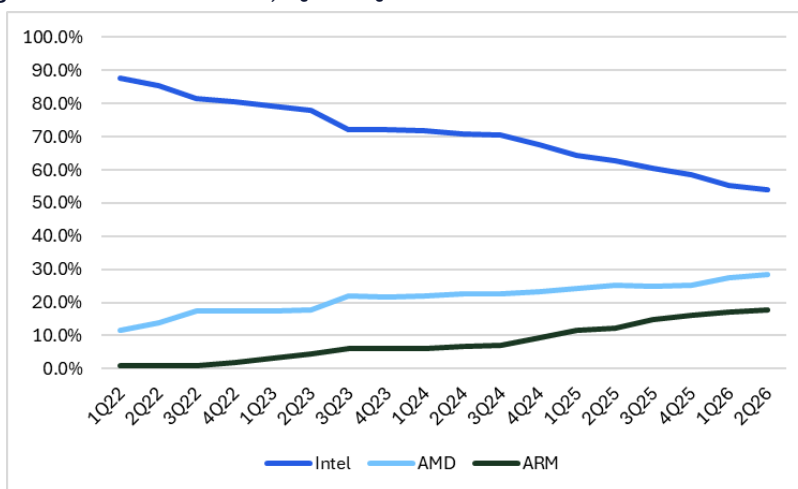
© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.

Source: Citi Research, Mercury Research

Market Share Update

Mercury Research released estimates for 2Q26 microprocessor shipments and market share. Intel's server MPU share was down 137 basis points QoQ from 55.3% in 1Q26 to 53.9% in 2Q26 while AMD's server MPU share was up 89 basis points QoQ from 27.5% in 1Q26 to 28.4% in 2Q26 due to our belief of better available CPU supply. Arm's server MPU share was up 48 basis points QoQ from 17.2% in 1Q26 to 17.6% in 2Q26.

Figure 7. Server MPU Share, 1Q22-2Q26

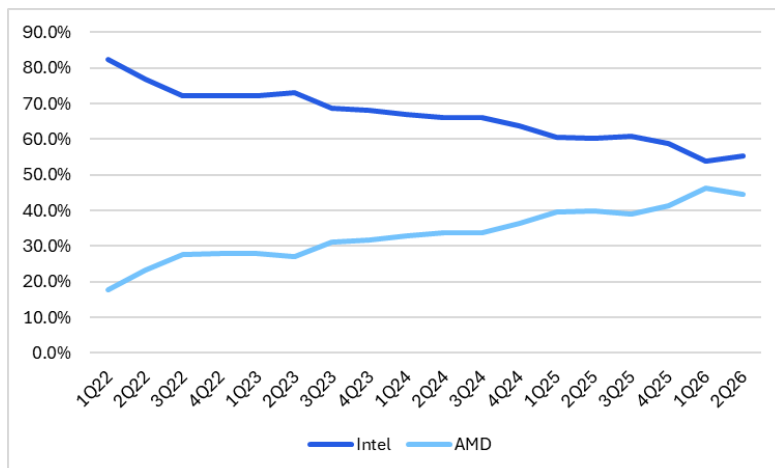


© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.

Source: Citi Research, Mercury Research

Server MPU revenue share, which excludes Arm, saw Intel gain 168 basis points QoQ from 53.8% in 1Q26 to 55.4% in 2Q26 on richer mix and the company raised ASPs. AMD's server revenue share declined 168 basis points QoQ from 46.2% in 1Q26 to 44.6% in 2Q26.

Figure 8. Server MPU Revenue Share, 1Q22-2Q26



© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.
Source: Citi Research, Mercury Research

Hot Chips CPU Notes:

We attended the Hot Chips conference recently, where we became increasingly constructive on the innovation occurring in the space. Please find the major CPU product details from Nvidia, Intel, Arm, and Fujitsu below:

Nvidia Vera CPU

- Agentic AI is the most complex computing workload in history. AI is moving from a query to a full workflow
- Vera is Nvidia's second generation server SOC and designed for agentic workflows
- Vera Rubin is a full-stack AI factory platform: Vera Rubin NVL-72 for training/inference, Groq 3 LPX, Vera CPU Rack, Vera Bluefield storage, and Spectrum X networking
- Vera Rubin NVL72 delivers up to a 30X throughput per MW versus GB300 on Deepseek
- LPX 3rd partner benchmark - 4x faster decode
- Vera CPU maximizes agent performance per core, max single threaded CPU at scale
- 5 important innovations on Vera that differentiates it from other DC CPUs: massive cores, built for power limited data centers, built to process massive data, completely protected, and massive IO
- Massive Cores: Olympus delivers more agent performance per core, per watt
- Built to process massive data: scalable coherency eliminates data ingestion bottleneck. Monolithic compute die, 88 Nvidia custom Olympus cores with 176 threads

- Vera delivers up to 6X more data processing performance versus a traditional CPU - in study with NYSE
- Massive I/O: Vera's massive I/O keeps GPU and Memory Investments fully utilized

Fujitsu- Arm based CPU

- Agentic AI drives greater CPU usage for orchestration and peripheral processing.
- Fujitsu-Monaka processor is made in Japan CPU engineered for AI performance and power efficiency
- Production shipments will begin in 2027
- Features up to 144 cores per socket, with options for both air and liquid cooling
- Uses 2nm
- Next CPU will launch in 2029 with a 1.4nm CPU for Data center, HPC, and AI

Arm AGI CPU

- Arm share is over 60% for host AI CPU platforms
- Arm AGI features TSMC N3P node and up to 136 cores (3 variants: 64 cores, 128 cores, and 136 cores)
- Each chiplet features up to 50B transistors, for over 100B transistors in a dual-transistor design
- Memory subsystem: up to 8800 MT/s on DDR5

Intel Diamond Rapids

- CPUs for data centers need higher performance per thread, maximize performance per watt, run the right workloads, need a scalable architecture, and need security. CPU needs to balance all these.
- Next Xeon architecture will be called Fan-out fabric architecture - which will have scalable compute building blocks, centralized I/O and memory, unified memory fabric, and a flexible I/O fabric
- Diamond Rapids is in the lab today
- Diamond Rapids SOC Overview: up to 256 cores total, 4 compute building blocks, 16 channel memory, 128 lanes of I/O, PCIe Gen 6, CXL 3.0, UPI 3
- Fabric Tile are built on Intel 3, Core is built on Intel 18A-P
- Intel 18A-P gives a 9% performance improvement over Intel 18A, and Intel 18A had a 15% performance improvement over the prior generation

Lifting 2026/27/28 WFE

Top-down WFE

We update our top-down WFE analysis for 2026/2027/2028 based on [Citi's updated hyperscaler capex model](#) of 106%/56%/33% growth in CY26/27/28. We

now see base case WFE of \$158Bn/\$209Bn/\$257Bn, with bull case WFE reaching \$163Bn/\$224Bn/\$286Bn in 2026/27/28 (including \$3Bn/\$10Bn/\$10Bn Terafab spending), up from \$147Bn/\$200Bn/\$250Bn prior and implies 37%/27% growth in both 2027/2028 in the bull case.

During the Jun-Q/July-Q earnings, KLAC, LRCX, and TEL all lifted 2026 WFE expectations to \$150Bn+, and AMAT also raised its semis sales expectations this year. KLAC explicitly noted that advanced packaging growth is now in the mid-30% range vs 20% prior, while LRCX, KLAC and AMAT all revised up their AP revenue to over 70% Y/Y vs 50% prior. Looking beyond 2026, **AMAT** sees CY27 to be another strong year with leading edge F/L, DRAM and AP to account for more than 80% of the growth, **Tokyo Electron** now expects CY27 WFE to be \$190Bn+ and can support \$200Bn-\$250Bn from CY27 onwards, **KLAC** endorsed industry consensus of \$190Bn WFE with upside potential, citing unprecedented visibility into 2H 2027 and even has early conversations for 2029 and beyond, and **LRCX** characterized demand environment as extraordinary setup for 2027 WFE growth (DRAM > leading edge F/L > NAND) and sees the favorable trend to continue into 2028, all driven by robust AI-related semiconductor demand, broad-based capacity additions across DRAM and logic, and increasing advanced packaging intensity. Lam now expects \$1-2Bn additional WFE spending on top of the \$8Bn prior from every \$100Bn data center capex. Against the backdrop of robust WFE growth, **TER** now expects ATE TAM to stabilize at 7-9% WFE driven by increasing chiplets complexity and higher test intensity requirements for AI processors and memory, reversing the trend of undergrowing in the last decade. **Advantest** revised up 2026 ATE TAM by 19% to grow ~50% Y/Y, supported by increasing volumes of AI related semis and device complexity.

Figure 9. WFE reaching \$209Bn/\$257Bn in CY27/28, with bull case of \$224Bn/\$286Bn.

Base												
WFE Mix	2025	2026 Growth	WFE 2026E	2027 Growth	WFE 2027E	2028 Growth	WFE 2028E	2028 WFE Mix	2029 Growth	WFE 2029E	2030 Growth	WFE 2030E
AI	\$ 45	106%	\$ 93	56%	\$ 144	33%	\$ 192	75%	36%	\$ 261	30%	\$ 338
China	\$ 39	0%	\$ 39	0%	\$ 39	0%	\$ 39	15%	0%	\$ 39	0%	\$ 39
Other Non-AI	\$ 27	0%	\$ 27	0%	\$ 27	0%	\$ 27	10%	0%	\$ 27	0%	\$ 27
Total WFE (\$Bn)	\$ 110	43%	\$ 158	33%	\$ 209	23%	\$ 257		27%	\$ 326	24%	\$ 403
Bull												
WFE Mix	2025	2026 Growth	WFE 2026E	2027 Growth	WFE 2027E	2028 Growth	WFE 2028E	2028 WFE Mix	2029 Growth	WFE 2029E	2030 Growth	WFE 2030E
AI	\$ 45	111%	\$ 95	61%	\$ 152	38%	\$ 210	74%	41%	\$ 296	35%	\$ 400
China	\$ 39	5%	\$ 40	5%	\$ 42	5%	\$ 45	16%	5%	\$ 47	5%	\$ 49
Other Non-AI	\$ 27	5%	\$ 28	5%	\$ 29	5%	\$ 31	11%	5%	\$ 32	5%	\$ 34
Total WFE (\$Bn)	\$ 110	48%	\$ 163	37%	\$ 224	27%	\$ 286		32%	\$ 376	29%	\$ 483

© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.

Source: Citi Research

Bottom-up WFE

We also update our bottom-up WFE model based on top spenders' capex comments for CY26 and beyond to align with our base case.

During the quarter, **TSMC** raised 2026 capex to \$60-64Bn and expects spending over the next three years to be significantly above the prior 3-year period, supported by advanced-node and packaging expansion in Taiwan and its enlarged \$265B Arizona plan. **Intel** raised 2026 capex to over \$20Bn, with tooling up roughly 40%, and expects 2027 capex to increase significantly again as it expands Intel 3, 18A/18A-P and advanced-packaging capacity, primarily in the U.S., alongside a €5B Ireland expansion. While **Samsung** has not provided precise 2026 guidance but expects a substantial increase from 2025's KRW52.7T, driven by Pyeongtaek and Taylor, while its long-term domestic plan includes KRW2,100T of semiconductor investment through 2040 across Pyeongtaek, Yongin, Gwangju and new HBM capacity. **SK Hynix** expects 2026 capex in the high-KRW40T range, led by M15X and the early-2027 opening of Yongin's first cleanroom; longer term, it

plans KRW1,100T across Yongin, Cheongju and a new southwestern Korea cluster, plus a \$4B+ Indiana HBM-packaging facility starting production in 2029. **Micron** expects FY26 capex of approximately \$27B and indicated FY27 spending above \$10B per quarter and full year could be above mid-40Bn level, although more than half of the increase is construction-related; capacity additions include Idaho, New York, Virginia, Taiwan, Japan and Singapore under its \$250B+ US investment plan. Finally, **Sandisk and Kioxia** plan to invest more than \$31B through 2032 in Yokkaichi and Kitakami, including a new Fab 3 targeted for FY2029. Overall, these plans support stronger WFE in 2027 and sustained tool installations through 2028–30, although the largest long-term commitments are phased project envelopes rather than near-term equipment orders.

Figure 10. Bottom-up WFE Model

Chn Top Spender Model				2026 CAPEX Outlook/WFE/GUIDE			26E	27E	28E
(US\$ millions)	23	24	25						
Global Foundries	\$1,804	\$625	\$722	Net CY26 capex in the range of 15% to 20% of full year revenue, trending toward higher end of the range			\$1,461	\$1,402	\$1,472
Intel	\$25,750	\$24,000	\$17,672	CY26 gross capex >\$20Bn, tooling 40% higher vs 2025; 2027 capex significantly above 2026			\$20,261	\$29,156	\$35,000
Rapiscap		\$0	\$5,000	FY25 budget of \$800Bn; FY26 budget of \$631Bn			\$4,000	\$4,000	\$4,000
Micron Technology	\$7,708	\$9,800	\$20,062	FY26 capex \$27Bn; FY27 up significantly, higher than mid \$40Bn			\$35,213	\$53,000	\$66,180
Nanya Technology	\$424	\$456	\$429	FY26 capex TWD 69.7 approved, 50% WFE; 2027 could be 4x			\$2,230	\$6,922	\$4,461
Sandisk/ Kioxia	\$2,200	\$800	\$1,792	FY27 capex to be "6% of revenue"			\$2,075	\$3,189	\$3,451
SK Hynix	\$6,232	\$11,100	\$19,250	CY26 capex to reach KRW high-40 trillion			\$34,265	\$49,684	\$78,004
Samsung	\$41,127	\$33,984	\$33,250	FY26: Expect Memory CapEx to increase considering the market outlook.			\$51,538	\$63,391	\$83,676
TSMC	\$30,450	\$29,800	\$41,000	CY26 capex of \$60Bn - \$64Bn: Split as 70-80% in advanced node, 10% in specialty node, and 10%-20% in advanced packaging, testing and mask making and others.			\$62,000	\$76,500	\$85,700
UMC	\$2,919	\$2,800	\$1,600	CY26 CapEx now \$2Bn; 2027 up low teens, board approval of \$5Bn over the next few years			\$2,000	\$2,240	\$2,509
Teralab				15% of SPCC AI Capex goes to Teralab			\$8,102	\$21,600	\$23,250
ASE				CY26 capex of \$10.5Bn, including \$6.5Bn for equipments			\$10,500	\$10,290	\$8,541
Amkor				2026 capex of \$2.5Bn - \$3Bn			\$2,750	\$2,338	\$1,917
China (domestic)	\$30,000	\$35,000	\$36,000	Giti models flat total domestic China capex spending in 2026			\$36,000	\$42,480	\$45,878
Top Spender Capex \$MM	\$148,614	\$148,368	\$178,777				\$269,144	\$365,604	\$429,882
WFE %	-4%	0%	79%				47%	37%	79%
Total Semi Capex \$MM	\$166,882	\$168,103	\$198,626				\$372,783	\$574,278	\$645,876
Wafer Fab Equipment \$MM	\$96,016	\$98,187	\$110,287				\$167,123	\$215,894	\$259,824
WFE %	0%	3%	77%				43%	27%	79%
Segment WFE Breakdown \$MM									
Logic (Foundry + IDM)	\$71,420	\$72,614	\$75,072				\$103,621	\$139,822	\$158,333
WFE %	73%	2%	3%				38%	35%	73%
Memory	\$29,262	\$26,554	\$35,185				\$53,592	\$75,763	\$88,492
WFE %	28%	24%	20%				15%	20%	20%
DRAM	\$17,917	\$19,003	\$25,874				\$41,398	\$60,028	\$78,037
WFE %	17%	16%	19%				10%	14%	14%
NAND	\$5,045	\$6,550	\$9,310				\$12,194	\$15,735	\$20,455
WFE %	5%	10%	26%				3%	3%	3%
Advanced Packaging							\$15,593	\$15,000	\$16,916
WFE %							4%	4%	7%
China WFE	\$35,000	\$41,225	\$44,393				\$47,103	\$50,851	\$55,912
WFE %		28%	2%				13%	14%	70%

© 2026 Citigroup Inc. No redistribution without Citigroup's written permission.

Source: Citi Research, Company Reports

Companies Mentioned:

Advanced Energy Industries, Inc (AEIS.O; US\$280.9; 1; 04 Sep 26; 16:00) |
Advanced Micro Devices (AMD.O; US\$477.57; 1; 04 Sep 26; 16:00) | Advantest
(6857.T; ¥34480.0; 1; 07 Sep 26; 15:30) | Amkor Technology, Inc. (AMKR.O;
US\$47.77; Not Rated; 04 Sep 26; 16:00) | Analog Devices (ADI.O; US\$362.25; 1; 04
Sep 26; 16:00) | Applied Materials Inc (AMAT.O; US\$454.71; 1; 04 Sep 26; 16:00) |
Arm Holdings PLC (ARM.O; US\$252.09; 1; 04 Sep 26; 16:00) | ASE Technology
Holding (3711.TW; NT\$621.0; 1; 07 Sep 26; 13:30) | ASML Holding NV (ASML.AS;
€1500.4; 1; 07 Sep 26; 16:30) | Astera Labs, Inc (ALAB.O; US\$310.4; 1; 04 Sep 26;
16:00) | Broadcom Inc (AVGO.O; US\$357.9; 1; 04 Sep 26; 16:00) | Cerebras Systems
Inc (CBRS.O; US\$210.05; 1; 04 Sep 26; 16:00) | Entegris Inc (ENTG.O; US\$138.74; 1;
04 Sep 26; 16:00) | FormFactor Inc (FORM.O; US\$103.9; 1; 04 Sep 26; 16:00) |
Fujitsu (6702.T; ¥3831.0; 1; 07 Sep 26; 15:30) | Global Foundries (GFS.O; US\$45.21;
2; 04 Sep 26; 16:00) | Infineon Technologies (IFXGn.DE; €60.79; 1; 07 Sep 26; 17:30)
| Intel Corp (INTC.O; US\$95.8; 1; 04 Sep 26; 16:00) | Kioxia Holdings (285A.T;
¥59530.0; 1H; 07 Sep 26; 15:30) | KLA Corp (KLAC.O; US\$185.6; 1; 04 Sep 26;
16:00) | Lam Research Corp (LRCX.O; US\$307.65; 1; 04 Sep 26; 16:00) | Lumentum
Holdings Inc (LITE.O; US\$881.26; 1; 04 Sep 26; 16:00) | Marvell Technology Inc

(MRVL.O; US\$223.55; 1; 04 Sep 26; 16:00) | Microchip Technology (MCHP.O; US\$74.17; 1; 04 Sep 26; 16:00) | Micron Technology Inc (MU.O; US\$1016.59; 1; 04 Sep 26; 16:00) | MKS Inc (MKSI.O; US\$260.31; 1; 04 Sep 26; 16:00) | Monolithic Power Systems (MPWR.O; US\$1223.86; 1; 04 Sep 26; 16:00) | Nanya Technology (2408.TW; NT\$517.0; 3; 07 Sep 26; 13:30) | Nova Ltd (NVMI.O; US\$372.1; 1; 04 Sep 26; 16:00) | NVIDIA Corp (NVDA.O; US\$230.36; 1; 04 Sep 26; 16:00) | NXP Semiconductors NV (NXPI.O; US\$227.84; 1; 04 Sep 26; 16:00) | ON Semiconductor (ON.O; US\$74.38; 2; 04 Sep 26; 16:00) | Qualcomm Inc (QCOM.O; US\$168.74; 2; 04 Sep 26; 16:00) | Samsung Electronics (005930.KS; W270000.O; 1; 07 Sep 26; 15:45) | Sandisk Corp (SNDK.O; US\$1740.0; 1; 04 Sep 26; 16:00) | Silicon Laboratories (SLAB.O; US\$220.58; 2; 04 Sep 26; 16:00) | SK Hynix (000660.KS; W1783000.O; 1; 07 Sep 26; 15:45) | Skyworks Solutions, Inc. (SWKS.O; US\$74.02; 2; 04 Sep 26; 16:00) | Space Exploration Technologies (SPCX.O; US\$147.95; 1; 04 Sep 26; 16:00) | STMicroelectronics (STMPA.PA; €46.64; 1; 07 Sep 26; 17:30) | Teradyne Inc (TER.O; US\$357.03; 1; 04 Sep 26; 16:00) | Texas Instruments Inc (TXN.O; US\$258.44; 1; 04 Sep 26; 16:00) | Tokyo Electron (8035.T; ¥55840.0; 1; 07 Sep 26; 15:30) | TSMC (2330.TW; NT\$2460.0; 1; 07 Sep 26; 13:30) | UMC (2303.TW; NT\$143.0; 1; 07 Sep 26; 13:30) | Veeco Instruments (VECO.O; US\$44.67; 1; 04 Sep 26; 16:00)

If you are visually impaired and would like to speak to a Citi representative regarding the details of the graphics in this document, please call USA 1-888-500-5008 (TTY: 711), from outside the US +1-210-677-3788

Appendix A-1

ANALYST CERTIFICATION

The research analysts primarily responsible for the preparation and content of this research report are either (i) designated by "AC" in the author block or (ii) listed in bold alongside content which is attributable to that analyst. If multiple AC analysts are designated in the author block, each analyst is certifying with respect to the entire research report other than (a) content attributable to another AC certifying analyst listed in bold alongside the content and (b) views expressed solely with respect to a specific issuer which are attributable to another AC certifying analyst identified in the price charts or rating history tables for that issuer shown below. Each of these analysts certify, with respect to the sections of the report for which they are responsible: (1) that the views expressed therein accurately reflect their personal views about each issuer and security referenced and were prepared in an independent manner, including with respect to Citigroup Global Markets Inc. and its affiliates; and (2) no part of the research analyst's compensation was, is, or will be, directly or indirectly, related to the specific recommendations or views expressed by that research analyst in this report.

IMPORTANT DISCLOSURES

The Reg AC analyst primarily responsible for this report is currently interviewing for a position with the company

A member of Citigroup's Board of Directors is also on the Board of Directors of Analog Devices.

An employee of Citigroup Global Markets Inc or its affiliates is a Director of Advanced Micro Devices, Inc.

James Bowlin, Senior Associate, used to be an officer, director or employee of Intel Corp.

Citigroup Global Markets Inc. or its affiliates has a net long position of 0.5% or more of any class of common equity securities of Astera Labs, Inc, Entegris Inc, Marvell Technology Inc, Microchip Technology, Monolithic Power Systems, NXP Semiconductors NV, Nova Ltd, ON Semiconductor, Skyworks Solutions, Inc..

Citigroup Global Markets Inc. or its affiliates has a net short position of 0.5% or more of any class of common equity securities of Kioxia Holdings, cs.

Within the past 12 months, Citigroup Global Markets Inc. or its affiliates has acted as manager or co-manager of an offering of securities of Advanced Energy Industries, Inc, Advanced Micro Devices, Applied Materials Inc, Broadcom Inc, Cerebras Systems Inc, Global Foundries, Infineon Technologies, Intel Corp, MKS Inc, Marvell Technology Inc, NVIDIA Corp, ON Semiconductor, SK Hynix, cs, Samsung Electronics, Sandisk Corp, Skyworks Solutions, Inc., Space Exploration Technologies.

Citigroup Global Markets Inc. or its affiliates has received compensation for investment banking services provided within the past 12 months from Advanced Energy Industries, Inc, Advanced Micro Devices, Analog Devices, Applied Materials Inc, Arm Holdings PLC, Broadcom Inc, Cerebras Systems Inc, Entegris Inc, Global Foundries, Infineon Technologies, Intel Corp, MKS Inc, Marvell Technology Inc, Micron Technology Inc, NVIDIA Corp, NXP Semiconductors NV, Nova Ltd, ON Semiconductor, Qualcomm Inc, SK Hynix, cs, Samsung Electronics, Sandisk Corp, Skyworks Solutions, Inc., Space Exploration Technologies, TSMC, Texas Instruments Inc, UMC.

Citigroup Global Markets Inc. or its affiliates expects to receive or intends to seek, within the next three months, compensation for investment banking services from Analog Devices, Arm Holdings PLC, Global Foundries, Intel Corp, KLA Corp, SK Hynix, Samsung Electronics, Sandisk Corp, Skyworks Solutions, Inc., Texas Instruments Inc, UMC.

Citigroup Global Markets Inc. or its affiliates received compensation for products and services other than investment banking services from ASE Technology Holding, ASML Holding NV, Advanced Energy Industries, Inc, Advanced Micro Devices, Advantest, Amkor Technology, Inc., Analog Devices, Applied Materials Inc, Arm Holdings PLC, Broadcom Inc, Cerebras Systems Inc, Entegris Inc, Fujitsu, Global Foundries, Infineon Technologies, Intel Corp, KLA Corp, Kioxia Holdings, Lam Research Corp, Lumentum Holdings Inc, MKS Inc, Marvell Technology Inc, Micron Technology Inc, Monolithic Power Systems, NVIDIA Corp, NXP Semiconductors NV, Nanya Technology, Nova Ltd, ON Semiconductor, Qualcomm Inc, SK Hynix, cs, Samsung Electronics, Sandisk Corp, Silicon Laboratories, Skyworks Solutions, Inc., Space Exploration Technologies, TSMC, Teradyne Inc, Texas Instruments Inc, Tokyo Electron, UMC, Veeco Instruments in the past 12 months.

Citigroup Global Markets Inc. or its affiliates currently has, or had within the past 12 months, the following as investment banking client(s): Advanced Energy Industries, Inc, Advanced Micro Devices, Analog Devices, Applied Materials Inc, Arm Holdings PLC, Broadcom Inc, Cerebras Systems Inc, Entegris Inc, Global Foundries, Infineon Technologies, Intel Corp, KLA Corp, MKS Inc, Marvell Technology Inc, Micron Technology Inc, NVIDIA Corp, NXP Semiconductors NV, Nova Ltd, ON Semiconductor, Qualcomm Inc, SK Hynix, cs, Samsung Electronics, Sandisk Corp, Skyworks Solutions, Inc., Space Exploration Technologies, TSMC, Texas Instruments Inc, UMC, Veeco Instruments.

Citigroup Global Markets Inc. or its affiliates currently has, or had within the past 12 months, the following as clients, and the services provided were non-investment-banking, securities-related: ASE Technology Holding, ASML Holding NV, Advanced Energy Industries, Inc, Advanced Micro Devices, Advantest, Amkor Technology, Inc., Analog Devices, Applied Materials Inc, Arm Holdings PLC, Broadcom Inc, Entegris Inc, Fujitsu, Global Foundries, Infineon Technologies, Intel Corp, KLA Corp, Kioxia

Holdings, Lam Research Corp, Lumentum Holdings Inc, MKS Inc, Marvell Technology Inc, Micron Technology Inc, Monolithic Power Systems, NVIDIA Corp, NXP Semiconductors NV, Nanya Technology, Nova Ltd, ON Semiconductor, Qualcomm Inc, SK Hynix, cs, Samsung Electronics, Sandisk Corp, Silicon Laboratories, Space Exploration Technologies, TSMC, Texas Instruments Inc, Tokyo Electron, UMC.

Citigroup Global Markets Inc. or its affiliates currently has, or had within the past 12 months, the following as clients, and the services provided were non-investment-banking, non-securities-related: ASE Technology Holding, ASML Holding NV, Advanced Energy Industries, Inc, Advanced Micro Devices, Advantest, Amkor Technology, Inc., Analog Devices, Applied Materials Inc, Arm Holdings PLC, Broadcom Inc, Cerebras Systems Inc, Entegris Inc, Fujitsu, Global Foundries, Infineon Technologies, Intel Corp, KLA Corp, Kioxia Holdings, Lam Research Corp, Lumentum Holdings Inc, MKS Inc, Marvell Technology Inc, Micron Technology Inc, Monolithic Power Systems, NVIDIA Corp, NXP Semiconductors NV, Nanya Technology, ON Semiconductor, Qualcomm Inc, SK Hynix, cs, Samsung Electronics, Sandisk Corp, Silicon Laboratories, Skyworks Solutions, Inc., Space Exploration Technologies, TSMC, Teradyne Inc, Texas Instruments Inc, Tokyo Electron, UMC, Veeco Instruments.

Citigroup Global Markets Inc. and/or its affiliates has a significant financial interest in relation to ASE Technology Holding, ASML Holding NV, Advanced Energy Industries, Inc, Advanced Micro Devices, Advantest, Amkor Technology, Inc., Analog Devices, Applied Materials Inc, Arm Holdings PLC, Broadcom Inc, Cerebras Systems Inc, Entegris Inc, Global Foundries, Infineon Technologies, Intel Corp, KLA Corp, Lam Research Corp, Lumentum Holdings Inc, MKS Inc, Marvell Technology Inc, Micron Technology Inc, NVIDIA Corp, NXP Semiconductors NV, ON Semiconductor, Qualcomm Inc, SK Hynix, cs, Samsung Electronics, Sandisk Corp, Silicon Laboratories, Skyworks Solutions, Inc., Space Exploration Technologies, TSMC, Texas Instruments Inc, Tokyo Electron, UMC, Veeco Instruments. (For an explanation of the determination of significant financial interest, please refer to the policy for managing conflicts of interest which can be found at www.citiVelocity.com.)

Analysts' compensation is determined by Citi Research management and Citigroup's senior management and is based upon activities and services intended to benefit the investor clients of Citigroup Global Markets Inc. and its affiliates (the "Firm"). Compensation is not linked to specific transactions or recommendations. Like all Firm employees, analysts receive compensation that is impacted by overall Firm profitability which includes investment banking, sales and trading, and principal trading revenues. One factor in equity research analyst compensation is arranging corporate access events between institutional clients and the management teams of covered companies. Typically, company management is more likely to participate when the analyst has a positive view of the company.

For financial instruments recommended in the Product in which the Firm is not a market maker, the Firm is a liquidity provider in such financial instruments (and any underlying instruments) and may act as principal in connection with transactions in such instruments. The Firm is a regular issuer of traded financial instruments linked to securities that may have been recommended in the Product. The Firm regularly trades in the securities of the issuer(s) discussed in the Product. The Firm may engage in securities transactions in a manner inconsistent with the Product and, with respect to securities covered by the Product, will buy or sell from customers on a principal basis.

Unless stated otherwise neither the Research Analyst nor any member of their team has viewed the material operations of the Companies for which an investment view has been provided within the past 12 months.

For important disclosures (including copies of historical disclosures) regarding the companies that are the subject of this Citi Research product ("the Product"), please contact Citi Research, 388 Greenwich Street, 6th Floor, New York, NY, 10013, Attention: Legal/Compliance [E6WYB6412478]. In addition, the same important disclosures, with the exception of the Valuation and Risk assessments and historical disclosures, are contained on the Firm's disclosure website at https://www.citivelocity.com/cvr/eppublic/citi_research_disclosures. Valuation and Risk assessments can be found in the text of the most recent research note/report regarding the subject company. Pursuant to the Market Abuse Regulation a history of all Citi Research recommendations published during the preceding 12-month period can be accessed via Citi Velocity (<https://www.citivelocity.com/cv2>) or your standard distribution portal. Historical disclosures (for up to the past three years) will be provided upon request.

Citi Research Equity Ratings Distribution

Data current as of 01 Jul 2026	12 Month Rating			Catalyst Watch		
	Buy	Hold	Sell	Buy	Hold	Sell
Citi Research Global Fundamental Coverage (Neutral=Hold)	61%	32%	7%	36%	48%	16%
% of companies in each rating category that are investment banking clients	38%	42%	27%	42%	37%	36%

Guide to Citi Research Fundamental Research Investment Ratings:

Citi Research stock recommendations include an investment rating and an optional risk rating to highlight high risk stocks.

Risk rating takes into account both price volatility and fundamental criteria. Stocks will either have no risk rating or a High risk rating assigned.

Investment Ratings: Citi Research investment ratings are Buy, Neutral and Sell. Our ratings are a function of analyst expectations of expected total return ("ETR") and risk. ETR is the sum of the forecast price appreciation (or depreciation) plus the dividend yield for a stock within the next 12 months. The target price is based on a 12 month time horizon. The Investment rating definitions are: Buy (1) ETR of 15% or more or 25% or more for High risk stocks; and Sell (3) for negative ETR. Any covered stock not assigned a Buy or a Sell is a Neutral (2). For stocks rated Neutral (2), if an analyst believes that there are insufficient valuation drivers and/or investment catalysts to derive a positive or negative investment view, they may elect with the approval

of Citi Research management not to assign a target price and, thus, not derive an ETR. Citi Research may suspend its rating and target price and assign "Rating Suspended" status for regulatory and/or internal policy reasons. Citi Research may also suspend its rating and target price and assign "Under Review" status for other exceptional circumstances (e.g. lack of information critical to the analyst's thesis, trading suspension) affecting the company and/or trading in the company's securities. In both such situations, the rating and target price will show as "—" and "—" respectively in the rating history price chart. Prior to 11 April 2022 Citi Research assigned "Under Review" status to both situations and prior to 11 Nov 2020 only in exceptional circumstances. As soon as practically possible, the analyst will publish a note re-establishing a rating and investment thesis. Investment ratings are determined by the ranges described above at the time of initiation of coverage, a change in investment and/or risk rating, or a change in target price (subject to limited management discretion). At times, the expected total returns may fall outside of these ranges because of market price movements and/or other short-term volatility or trading patterns. Such interim deviations will be permitted but will become subject to review by Research Management. Your decision to buy or sell a security should be based upon your personal investment objectives and should be made only after evaluating the stock's expected performance and risk.

Catalyst Watch/Short Term Views ("STV") Ratings Disclosure:

Catalyst Watch and STV Upside/Downside calls: Citi Research may also include a Catalyst Watch or STV Upside or Downside call to indicate the analyst expects the share price to rise (fall) in absolute terms over a specified period of 30 or 90 days in reaction to one or more specific near-term catalysts or events impacting the company or the market. A Catalyst Watch will be published when Analyst confidence is high that an impact to share price will occur; it will be a STV when confidence level is moderate. A Catalyst Watch or STV Upside/Downside call will automatically expire at the end of the specified 30/90 day period. The Catalyst Watch will also be automatically removed if share price performance (calculated at market close) exceeds 15% against the direction of the call (unless over-ridden by the analyst). The analyst may also remove a Catalyst Watch or STV call prior to the end of the specified period in a published research note. A Catalyst Watch/STV Upside or Downside call may be different from and does not affect a stock's fundamental equity rating, which reflects a longer-term total absolute return expectation. For purposes of FINRA ratings-distribution-disclosure rules, a Catalyst Watch/STV Upside call corresponds to a buy recommendation and a Catalyst Watch/STV Downside call corresponds to a sell recommendation. Any stock not assigned to a Catalyst Watch Upside, Catalyst Watch Downside, STV Upside, or STV Downside call is considered Catalyst Watch/STV No View. For purposes of FINRA ratings distribution-disclosure rules, we correspond Catalyst Watch/STV No View to Hold in our ratings distribution table for our Catalyst Watch/STV Upside/Downside rating system. However, we reiterate that we do not consider No View to be a recommendation. For all Catalyst Watch/STV Upside/Downside calls, risk exists that the catalyst(s) and associated share-price movement will not materialize as expected.

Privately held companies:

The legal entities through which Citi conducts its business do not permit investment research recommendations, ratings, target prices, or estimates on any privately held companies ("private companies") referenced herein. Citi Research does not utilize a rating system for private companies. Any information on private companies in this report is based on generally available information and, although obtained from sources believed by Citi to be reliable, its accuracy and completeness are not guaranteed. Private companies are generally not subject to reporting obligations, which may result in less publicly available information. The information contained in this report may differ from information a private company may subsequently make public or otherwise share with investors. Continuation of publishing private company reports or private company sections within reports is at Citi's discretion, in accordance with internal policies.

RESEARCH ANALYST AFFILIATIONS / NON-US RESEARCH ANALYST DISCLOSURES

The legal entities employing the authors of this report are listed below (and their regulators are listed further herein). Non-US research analysts who have prepared this report (i.e., all research analysts listed below other than those identified as employed by Citigroup Global Markets Inc.) are not registered/qualified as research analysts with FINRA. Such research analysts may not be associated persons of the member organization (but are employed by an affiliate of the member organization) and therefore may not be subject to the FINRA Rule 2241 restrictions on communications with a subject company, public appearances and trading securities held by a research analyst account.

Citigroup Global Markets Inc.

James Bowlin; Papa Sylla; Atif Malik; Elizabeth Sun, CFA; Kelsey Chia, CFA

OTHER DISCLOSURES

Any price(s) of instruments mentioned in recommendations are as of the prior day's market close on the primary market for the instrument, unless otherwise stated.

The completion and first dissemination of any recommendations made within this research report are as of the Eastern date-time displayed at the top of the Product. If the Product references views of other analysts then please refer to the price chart or rating history table for the date/time of completion and first dissemination with respect to that view.

Regulations in various jurisdictions require that where a recommendation differs from any of the author's previous recommendations concerning the same financial instrument or issuer that has been published during the preceding 12-month period that the change(s) and the date of that previous recommendation are indicated. For fundamental coverage please refer to the price chart or rating change history within this disclosure appendix or the issuer disclosure summary at https://www.citivelocity.com/cvr/eppublic/citi_research_disclosures.

Citi Research has implemented policies for identifying, considering and managing potential conflicts of interest arising as a result of publication or distribution of investment research. A description of these policies can be found at https://www.citivelocity.com/cvr/eppublic/citi_research_disclosures.

The proportion of all Citi Research research recommendations that were the equivalent to "Buy", "Hold", "Sell" at the end of each quarter over the prior 12 months (with the % of these that had received investment firm services from Citi in the prior 12 months shown in brackets) is as follows; Q2 2026 Buy 33%(63%), Hold 44% (51%), Sell 23% (49%), RV 0.5% (87%); Q1 2026 Buy 33% (63%), Hold 44% (52%), Sell 23% (46%), RV 0.5% (89%); Q4 2025 Buy 33% (63%), Hold 44% (50%), Sell 23% (46%), RV 0.4% (91%); Q3 2025 Buy 33%(61%), Hold 44% (52%), Sell 23% (50%), RV 0.4% (80%). For the purposes of disclosing recommendations other than for equity (whose definitions can be found in the corresponding disclosure sections), "Buy" means a positive directional trade idea; "Sell" means a negative directional trade idea; and "Relative Value" means any trade idea which does not have a clear direction to the investment strategy.

European regulations require a 5 year price history when past performance of a security is referenced. CitiVelocity's Charting Tool (https://www.citivelocity.com/cv2/#go/CHARTING_3_Equities) provides the facility to create customisable price charts including a five year option. This tool can be found in the Data & Analytics section under any of the asset class menus in CitiVelocity (<https://www.citivelocity.com/>). For further information contact CitiVelocity support (https://www.citivelocity.com/cv2/go/CLIENT_SUPPORT). The source for all referenced prices, unless otherwise stated, is DataCentral, which sources price information from LSEG Data & Analytics. Past performance is not a guarantee or reliable indicator of future results. Forecasts are not a guarantee or reliable indicator of future performance.

Investors should always consider the investment objectives, risks, and charges and expenses of an ETF carefully before investing. The applicable prospectus and key investor information document (as applicable) for an ETF should contain this and other information about such ETF. It is important to read carefully any such prospectus before investing. Clients may obtain prospectuses and key investor information documents for ETFs from the applicable distributor or authorized participant, the exchange upon which an ETF is listed and/or from the applicable website of the applicable ETF issuer. The value of the investments and any accruing income may fall or rise. Any past performance, prediction or forecast is not indicative of future or likely performance. Any information on ETFs contained herein is provided strictly for illustrative purposes and should not be deemed an offer to sell or a solicitation of an offer to purchase units of any ETF either explicitly or implicitly. The opinions expressed are those of the authors and do not necessarily reflect the views of ETF issuers, any of their agents or their affiliates.

Citigroup Global Markets India Private Limited and/or its affiliates may have, from time to time, actual or beneficial ownership of 1% or more in the debt securities of the subject issuer.

Please be advised that pursuant to Executive Order 13959 as amended (the "Order"), U.S. persons are prohibited from investing in securities of any company determined by the United States Government to be the subject of the Order. This research is not intended to be used or relied upon in any way that could result in a violation of the Order. Investors are encouraged to rely upon their own legal counsel for advice on compliance with the Order and other economic sanctions programs administered and enforced by the Office of Foreign Assets Control of the U.S. Treasury Department.

This communication is directed at persons who are "Eligible Clients" as such term is defined in the Israeli Regulation of Investment Advice, Investment Marketing and Investment Portfolio Management law, 1995 (the "Advisory Law"). Within Israel, this communication is not intended for retail clients and Citi will not make such products or transactions available to retail clients or to non-Eligible Clients. The presenter is not licensed as investment advisor or investment marketer by the Israeli Securities Authority ("ISA") and this communication does not constitute investment or marketing advice. The information contained herein may relate to matters that are not regulated by the ISA. Any securities which are the subject of this communication may not be offered or sold to any Israeli person except pursuant to a security offering exemption according to the Israeli Securities Law, 1968 and the public offering rules provided thereunder.

Citi Research broadly and simultaneously disseminates its research content to the Firm's institutional and retail clients via the Firm's proprietary electronic distribution platforms (e.g., Citi Velocity and various Global Wealth platforms). As a convenience, certain, but not all, research content may be distributed through third party aggregators. Clients may receive published research reports by email, on a discretionary basis, and only after such research content has been broadly disseminated. Certain research is made available only to institutional investors to satisfy regulatory requirements. The level and types of services provided by Citi Research analysts to clients may vary depending on various factors such as the client's individual preferences as to the frequency and manner of receiving communications from analysts, the client's risk profile and investment focus and perspective (e.g. market-wide, sector specific, long term, short-term etc.), the size and scope of the overall client relationship with the Firm and legal and regulatory constraints.

Pursuant to Comissão de Valores Mobiliários Resolução 20 and ASIC Regulatory Guide 264, Citi is required to disclose whether a Citi related company or business has a commercial relationship with the subject company. Considering that Citi operates multiple businesses in more than 100 countries around the world, it is likely that Citi has a commercial relationship with the subject company.

Securities recommended, offered, or sold by the Firm: (i) are not insured by the Federal Deposit Insurance Corporation; (ii) are not deposits or other obligations of any insured depository institution (including Citibank); and (iii) are subject to investment risks, including the possible loss of the principal amount invested. The Product is for informational purposes only and is not intended as an offer or solicitation for the purchase or sale of a security. Any decision to purchase securities mentioned in the Product must take into account existing public information on such security or any registered prospectus. Although information has been obtained from and is based upon sources that the Firm believes to be reliable, we do not guarantee its accuracy and it may be incomplete and condensed. Note, however, that the Firm has taken all reasonable steps to determine

the accuracy and completeness of the disclosures made in the Important Disclosures section of the Product. The Firm's research department has received assistance from the subject company(ies) referred to in this Product including, but not limited to, discussions with management of the subject company(ies). Firm policy prohibits research analysts from sending draft research to subject companies. However, it should be presumed that the author of the Product has had discussions with the subject company to ensure factual accuracy prior to publication. Statements and views concerning ESG (environmental, social, governance) factors are typically based upon public statements made by the affected company or other public news, which the author may not have independently verified. ESG factors are one consideration that investors may choose to examine when making investment decisions. All opinions, projections and estimates constitute the judgment of the author as of the date of the Product and these, plus any other information contained in the Product, are subject to change without notice. Prices and availability of financial instruments also are subject to change without notice. Notwithstanding other departments within the Firm advising the companies discussed in this Product, information obtained in such role is not used in the preparation of the Product. Although Citi Research does not set a predetermined frequency for publication, if the Product is a fundamental equity or credit research report, it is the intention of Citi Research to provide research coverage of the covered issuers, including in response to news affecting the issuer. For non-fundamental research reports, Citi Research may not provide regular updates to the views, recommendations and facts included in the reports. Notwithstanding that Citi Research maintains coverage on, makes recommendations concerning or discusses issuers, Citi Research may be periodically restricted from referencing certain issuers due to legal or policy reasons. Where a component of a published trade idea is subject to a restriction, the trade idea will be removed from any list of open trade ideas included in the Product. Upon the lifting of the restriction, the trade idea will either be re-instated in the open trade ideas list if the analyst continues to support it or it will be officially closed. Citi Research may provide different research products and services to different classes of customers (for example, based upon long-term or short-term investment horizons) that may lead to differing conclusions or recommendations that could impact the price of a security contrary to the recommendations in the alternative research product, provided that each is consistent with the rating system for each respective product.

Investing in non-U.S. securities, including ADRs, may entail certain risks. The securities of non-U.S. issuers may not be registered with, nor be subject to the reporting requirements of the U.S. Securities and Exchange Commission. There may be limited information available on foreign securities. Foreign companies are generally not subject to uniform audit and reporting standards, practices and requirements comparable to those in the U.S. Securities of some foreign companies may be less liquid and their prices more volatile than securities of comparable U.S. companies. In addition, exchange rate movements may have an adverse effect on the value of an investment in a foreign stock and its corresponding dividend payment for U.S. investors. Net dividends to ADR investors are estimated, using withholding tax rates conventions, deemed accurate, but investors are urged to consult their tax advisor for exact dividend computations. Investors who have received the Product from the Firm may be prohibited in certain states or other jurisdictions from purchasing securities mentioned in the Product from the Firm. Please ask your Financial Consultant for additional details. Citigroup Global Markets Inc. takes responsibility for the Product in the United States. Any orders by US investors resulting from the information contained in the Product may be placed only through Citigroup Global Markets Inc.

The Citigroup legal entity that takes responsibility for the production of the Product is the legal entity which the first named author is employed by.

The Product is made available in **Australia** through Citigroup Global Markets Australia Pty Limited. (ABN 64 003 114 832 and AFSL No. 240992), participant of the ASX Group and regulated by the Australian Securities & Investments Commission. Citigroup Centre, 2 Park Street, Sydney, NSW 2000. Citigroup Global Markets Australia Pty Limited is not an Authorised Deposit-Taking Institution under the Banking Act 1959, nor is it regulated by the Australian Prudential Regulation Authority.

The Product is made available in **Brazil** by Citigroup Global Markets Brasil - CCTVM SA, which is regulated by CVM - Comissão de Valores Mobiliários ("CVM"), BACEN - Brazilian Central Bank, APIMEC - Associação dos Analistas e Profissionais de Investimento do Mercado de Capitais and ANBIMA - Associação Brasileira das Entidades dos Mercados Financeiro e de Capitais. Av. Paulista, 1111 - 14º andar (parte) - CEP: 01311-920 - São Paulo - SP.

This Product is available in **Chile** through Banchile Corredores de Bolsa S.A., an indirect subsidiary of Citigroup Inc., which is regulated by the Comisión Para El Mercado Financiero. Enrique Foster Sur, 20, piso 6, Las Condes, Santiago, Chile.

Disclosure for investors in the Republic of Colombia : This communication or message does not constitute a professional recommendation to make investment in the terms of article 2.40.1.1.2 of Decree 2555 de 2010 or the regulations that modify, substitute or complement it. Para la elaboración y distribución de informes de investigación y de comunicaciones generales de que trata este artículo no se requiere ser una entidad vigilada por la Financiera de Colombia.

The Product is made available in **Germany** by Citigroup Global Markets Europe AG ("CGME"), which is regulated by the European Central Bank and the German Federal Financial Supervisory Authority (Bundesanstalt für
tunsaufsicht BaFin). Börsenplatz 9, 60313 Frankfurt am Main, Germany.

Unless otherwise specified, if the analyst who prepared this report is based in Hong Kong and it relates to "securities" (as defined in the Securities and Futures Ordinance (Cap.571 of the Laws of Hong Kong)), the report is issued in Hong Kong by Citigroup Global Markets Asia Limited. Citigroup Global Markets Asia Limited is regulated by Hong Kong Securities and Futures Commission. If the report is prepared by a non-Hong Kong based analyst, please note that such analyst (and the legal entity that the analyst is employed by or accredited to) is not licensed/registered in Hong Kong and they do not hold themselves out as such. Please refer to the section "Research Analyst Affiliations / Non-US Research Analyst Disclosures" for the details of the employment entity of the analysts.

The Product is made available in **India** by Citigroup Global Markets India Private Limited (CGM), which is regulated by the Securities and Exchange Board of India (SEBI), as a Research Analyst (SEBI Registration No. INH000000438). CGM is also actively involved in the business of merchant banking (SEBI Registration No. INM000010718) and stock brokerage ((SEBI Registration No. INZ000263033) in India, and is registered with SEBI in this regard. Registration granted by SEBI, enlistment with Bombay Stock Exchange (BSE) and certification from National Institute of Securities Markets (NISM) in no way guarantee performance of the intermediary or provide any assurance of returns to investors. CGM's registered office is at 1202, 12th Floor, First International Financial Centre (FIFC), G Block, Bandra Kurla Complex, Bandra East, Mumbai – 400098 & registered Tel: +91 22 61759999. Citi maintains robust policies, procedures, controls, and training to ensure continued compliance with all applicable rules and regulations. All recommendations contained herein are made by duly qualified research analysts. CGM's Corporate Identity Number is 26657, and its Compliance Officer [Vishal Bohra] contact details are: Tel: +91-022-61759994, Fax: +91-022-61759851, Email: cgmcompliance@citi.com. The Investor Charter in respect of Research Analysts, the Compliance Audit Report and Complaints information can be found at https://www.citivelocity.com/cvr/eppublic/citi_research_disclosures. The grievance officer [Niraj Mody] contact details are Tel: +91-22-42775002, Email: EMEA.CR.Complaints@citi.com. Investment in securities market are subject to market risks. Read all the related documents carefully before investing. SEBI prescribed Client Terms & Conditions can be found at https://www.citivelocity.com/rendition/_/eppublic/V1/file?paramData=L0dETVMvcHVjVXNlMh0bWw

The Product is made available in **Indonesia** through PT Citigroup Sekuritas Indonesia. Citibank Tower 10/F, Pacific Century Place, SCBD lot 10, Jl. Jend Sudirman Kav 52-53, Jakarta 12190, Indonesia. Neither this Product nor any copy hereof may be distributed in Indonesia or to any Indonesian citizens wherever they are domiciled or to Indonesian residents except in compliance with applicable capital market laws and regulations. This Product is not an offer of securities in Indonesia. The securities referred to in this Product have not been registered with the Capital Market and Financial Services Authority (OJK) pursuant to relevant capital market laws and regulations, and may not be offered or sold within the territory of the Republic of Indonesia or to Indonesian citizens through a public offering or in circumstances which constitute an offer within the meaning of the Indonesian capital market laws and regulations.

The Product is made available in **Japan** by Citigroup Global Markets Japan Inc. ("CGMJ"), which is regulated by Financial Services Agency, Securities and Exchange Surveillance Commission, Japan Securities Dealers Association, Tokyo Stock Exchange and Osaka Securities Exchange. Otemachi Park Building, 1-1-1 Otemachi, Chiyoda-ku, Tokyo 100-8132 Japan. In the event that an error is found in an CGMJ research report, a revised version will be posted on the Firm's Citi Velocity website. If you have questions regarding Citi Velocity, please call (81 3) 6270-3019 for help.

The product is made available in the **Kingdom of Saudi Arabia** in accordance with Saudi laws through Citigroup Saudi Arabia, which is regulated by the Capital Market Authority (CMA) under CMA license (17184-31). 2239 Al Urubah Rd – Al Olaya Dist. Unit No. 18, Riyadh 12214 – 9597, Kingdom Of Saudi Arabia.

The Product is made available in **Korea** by Citigroup Global Markets Korea Securities Ltd. (CGMK), which is regulated by the Financial Services Commission, the Financial Supervisory Service and the Korea Financial Investment Association (KOFIA). The address of CGMK is Citibank Center, 50 Saemunan-ro, Jongno-gu, Seoul 03184, Korea. KOFIA makes available registration information of research analysts on its website. Please visit the following website if you wish to find KOFIA registration information on research analysts of

CGMK. <http://dis.kofia.or.kr/websquare/index.jsp?w2xPath=/wq/fundMgr/tList.xml&divisionId=MDISO3002002000000&serviceld=SDISO3002002000>. The Product is made available in Korea by Citibank Korea Inc., which is regulated by the Financial Services Commission and the Financial Supervisory Service. Address is Citibank Center, 50 Saemunan-ro, Jongno-gu, Seoul 03184, Korea. This research report is intended to be provided only to Professional Investors as defined in the Financial Investment Services and Capital Market Act and its Enforcement Decree in Korea.

The Product is made available in **Malaysia** by Citigroup Global Markets Malaysia Sdn Bhd (Registration No. 199801004692 (460819-D)) ("CGMM") to its clients and CGMM takes responsibility for its contents as regards CGMM's clients. CGMM is regulated by the Securities Commission Malaysia. Please contact CGMM at Level 43 Menara Citibank, 165 Jalan Ampang, 50450 Kuala Lumpur, Malaysia in respect of any matters arising from, or in connection with, the Product.

The Product is made available in **Mexico** by Citi México Casa de Bolsa, S.A. de C.V., Grupo Financiero Citi México which is a wholly owned subsidiary of Citigroup Inc. and is regulated by Comisión Nacional Bancaria y de Valores. Prolongación Reforma 1196, 24 floor, Colonia Santa Fe, Alcaldía Cuajimalpa de Morelos, C.P. 05348, Ciudad de México.

The Product is made available in **Poland** by Biuro Maklerskie Banku Handlowego (DMBH), separate department of Bank Handlowy w Warszawie S.A. a subsidiary of Citigroup Inc., which is regulated by Komisja Nadzoru Finansowego. Biuro Maklerskie Banku Handlowego (DMBH), ul. Senatorska 16, 00-923 Warszawa.

The Product is made available in **Singapore** through Citigroup Global Markets Singapore Pte. Ltd. ("CGMSPL"), a capital markets services and Exempt Financial Advisor license holder, and regulated by Monetary Authority of Singapore. Please contact CGMSPL at 8 Marina View, 21st Floor Asia Square Tower 1, Singapore 018960, in respect of any matters arising from, or in connection with, the analysis of this document. This Product is intended for recipients who are accredited, expert and institutional investors as defined under the Securities and Futures Act 2001. For Citi Private Bank, the Product is made available in Singapore by Citi Private Bank through Citibank, N.A., Singapore Branch. Citibank N.A., Singapore Branch is a licensed bank in Singapore that is regulated by the Monetary Authority of Singapore. Please contact your Private Banker in Citibank N.A., Singapore Branch if you have any queries on or any matters arising from or in connection with this document. The Product is intended for recipients who are accredited, expert and institutional investors as defined under the Securities

and Futures Act 2001. For Citibank Singapore Limited ("CSL"), the Product is distributed in Singapore by CSL to selected Citigold/Citigold Private Clients. CSL provides no independent research or analysis of the substance or in preparation of the Product. Please contact your Citigold/Citigold Private Client Relationship Manager in CSL if you have any queries on or any matters arising from or in connection with this document. The Product is intended for recipients who are accredited investors as defined under the Securities and Futures Act (Cap. 289).

Citigroup Global Markets (Pty) Ltd. is incorporated in the **Republic of South Africa** (company registration number 2000/025866/07) and its registered office is at 145 West Street, Sandton, 2196, Saxonwold. Citigroup Global Markets (Pty) Ltd. is regulated by JSE Securities Exchange South Africa, South African Reserve Bank and the Financial Services Board. The investments and services contained herein are not available to private customers in South Africa.

The Product is made available in the **Republic of China (Taiwan)** through Citigroup Global Markets Taiwan Securities Company Ltd. ("CGMTS"), 14F, 15F and 16F, No. 1, Songzhi Road, Taipei 110, Taiwan, subject to the license scope and the applicable laws and regulations in the Republic of China (Taiwan). CGMTS is regulated by the Securities and Futures Bureau of the Financial Supervisory Commission of Taiwan, the Republic of China (Taiwan). No portion of the Product may be reproduced or quoted in the Republic of China (Taiwan) by the press or any third parties [without the written authorization of CGMTS]. Pursuant to the applicable laws and regulations in the Republic of China (Taiwan), the recipient of the Product shall not take advantage of such Product to involve in any matters in which the recipient may have conflicts of interest. If the Product covers securities which are not allowed to be offered or traded in the Republic of China (Taiwan), neither the Product nor any information contained in the Product shall be considered as advertising the securities or making recommendation of the securities in the Republic of China (Taiwan). The Product is for informational purposes only and is not intended as an offer or solicitation for the purchase or sale of a security or financial products. Any decision to purchase securities or financial products mentioned in the Product must take into account existing public information on such security or the financial products or any registered prospectus.

The Product is made available in **Thailand** through Citicorp Securities (Thailand) Ltd., which is regulated by the Securities and Exchange Commission of Thailand. 399 Interchange 21 Building, 18th Floor, Sukhumvit Road, Klongtoey Nua, Wattana, Bangkok 10110, Thailand.

Disclosure for investors in the Republic of Turkey: The Product is made available in **Turkey** through Citibank AS which is regulated by Capital Markets Board. Tekfen Tower, Eski Buyukdere Caddesi # 209 Kat 2B, 23294 Levent, Istanbul, Turkey. Under Capital Markets Law of Turkey (Law No: 6362), the investment information, comments and recommendations stated here, are not within the scope of investment advisory activity. Comments and recommendations stated here rely on the individual opinions of the ones providing these comments and recommendations. These opinions may not fit to your financial status, risk and return preferences. For this reason, to make an investment decision by relying solely to this information stated here may not bring about outcomes that fit your expectations. Furthermore, Citi Research is a division of Citigroup Global Markets Inc. (the "Firm"), which does and seeks to do business with companies and/or trades on securities covered in this research reports. As a result, investors should be aware that the Firm may have a conflict of interest that could affect the objectivity of this report, however investors should also note that the Firm has in place organisational and administrative arrangements to manage potential conflicts of interest of this nature.

In the **U.A.E.**, these materials (the "Materials") are communicated by Citigroup Global Markets Limited, DIFC branch ("CGML"), an in the Dubai International Financial Center ("DIFC") and licensed and regulated by the Dubai Financial Services Authority ("DFSA", license #CLO221) to Professional Clients and Market Counterparties, as defined in DFSA regulations, only and should not be relied upon or distributed to Retail Clients. Financial products and/or services to which the Materials relate will only be made available to Professional Clients and ties. Citigroup Global Markets Limited DIFC Branch registered address is Level 3, Gate District Building 02, Dubai International Financial Centre and can be contacted on +971 4 509 97 90.

The Product is made available in **United Kingdom** by Citigroup Global Markets Limited, which is authorised by the Prudential Regulation Authority ("PRA") and regulated by the Financial Conduct Authority ("FCA") and the PRA. This material may relate to investments or services of a person outside of the UK or to other matters which are not authorised by the PRA nor regulated by the FCA and the PRA and further details as to where this may be the case are available upon request in respect of this material. Citigroup Centre, Canada Square, Canary Wharf, London, E14 5LB.

The Product is made available in **United States** and **Canada** by Citigroup Global Markets Inc., which is a member of FINRA and registered with the US Securities and Exchange Commission. 388 Greenwich Street, New York, NY 10013.

Unless specified to the contrary, within EU Member States, the Product is made available by Citigroup Global Markets Europe AG ("CGME"), which is regulated by the European Central Bank and the German Federal Financial Supervisory Authority (Bundesanstalt für tungsaufsicht-BaFin).

The Product is not to be construed as providing investment services in any jurisdiction where the provision of such services would not be permitted.

Subject to the nature and contents of the Product, the investments described therein are subject to fluctuations in price and/or value and investors may get back less than originally invested. Certain high-volatility investments can be subject to sudden and large falls in value that could equal or exceed the amount invested. The yield and average life of CMOs (collateralized mortgage obligations) referenced in this Product will fluctuate depending on the actual rate at which mortgage holders prepay the mortgages underlying the CMO and changes in current interest rates. Any government agency backing of the CMO applies only to the face value of the CMO and not to any premium paid. Certain investments contained in the Product may have tax implications for private customers whereby levels and basis of taxation may be subject to change. If in doubt,

investors should seek advice from a tax adviser. The Product does not purport to identify the nature of the specific market or other risks associated with a particular transaction. Advice in the Product is general and should not be construed as personal advice given it has been prepared without taking account of the objectives, financial situation or needs of any particular investor. Accordingly, investors should, before acting on the advice, consider the appropriateness of the advice, having regard to their objectives, financial situation and needs. Prior to acquiring any financial product, it is the client's responsibility to obtain the relevant offer document for the product and consider it before making a decision as to whether to purchase the product.

Card Insights. Where this report references Card Insights data, Card Insights consists of selected data from a subset of Citi's proprietary credit card transactions. Such data has undergone rigorous security protocols to keep all customer information confidential and secure; the data is highly aggregated and anonymized so that all unique customer identifiable information is removed from the data prior to receipt by the report's author or distribution to external parties. This data should be considered in the context of other economic indicators and publicly available information. Further, the selected data represents only a subset of Citi's proprietary credit card transactions due to the selection methodology or other limitations and should not be considered as indicative or predictive of the past or future financial performance of Citi or its credit card business.

Citi Research product may source data from dataCentral. dataCentral is a Citi Research proprietary database, which includes the Firm's estimates, data from company reports and feeds from LSEG Data & Analytics. The source for all referenced prices, unless otherwise stated, is DataCentral. Past performance is not a guarantee or reliable indicator of future results. Forecasts are not a guarantee or reliable indicator of future performance. The printed and printable version of the research report may not include all the information (e.g. certain financial summary information and comparable company data) that is linked to the online version available on the Firm's proprietary electronic distribution platforms.

Where included in this report, MSCI sourced information is the exclusive property of Morgan Stanley Capital International Inc. (MSCI). Without prior written permission of MSCI, this information and any other MSCI intellectual property may not be reproduced, disseminated or used to create any financial products, including any indices. This information is provided on an "as is" basis. The user assumes the entire risk of any use made of this information. MSCI, its affiliates and any third party involved in, or related to, computing or compiling the information hereby expressly disclaim all warranties of originality, accuracy, completeness, merchantability or fitness for a particular purpose with respect to any of this information. Without limiting any of the foregoing, in no event shall MSCI, any of its affiliates or any third party involved in, or related to, computing or compiling the information have any liability for any damages of any kind. MSCI, Morgan Stanley Capital International and the MSCI indexes are services marks of MSCI and its affiliates. Where data is attributed to Morningstar that data is © 2026 Morningstar, Inc. All Rights Reserved. That information: (1) is proprietary to Morningstar and/or its content providers; (2) may not be copied or distributed; and (3) is not warranted to be accurate, complete or timely. Neither Morningstar nor its content providers are responsible for any damages or losses arising from any use of this information.

The Firm accepts no liability whatsoever for the actions of third parties. The Product may provide the addresses of, or contain hyperlinks to, websites. Except to the extent to which the Product refers to website material of the Firm, the Firm has not reviewed the linked site. Equally, except to the extent to which the Product refers to website material of the Firm, the Firm takes no responsibility for, and makes no representations or warranties whatsoever as to, the data and information contained therein. Such address or hyperlink (including addresses or hyperlinks to website material of the Firm) is provided solely for your convenience and information and the content of the linked site does not in any way form part of this document. Accessing such website or following such link through the Product or the website of the Firm shall be at your own risk and the Firm shall have no liability arising out of, or in connection with, any such referenced website.

© 2026 Citigroup Global Markets Inc. Citi Research is a division of Citigroup Global Markets Inc. Citi and Citi and Arc Design are trademarks and service marks of Citigroup Inc. and its affiliates and are used and registered throughout the world. All rights reserved. The research data in this report are not intended to be used for the purpose of (a) determining the price of or amounts due in respect of (or to value) one or more financial products or instruments and/or (b) measuring or comparing the performance of, or defining the asset allocation of a financial product, a portfolio of financial instruments, or a collective investment undertaking, and any such use is strictly prohibited without the prior written consent of Citi Research. Any unauthorized use, duplication, redistribution or disclosure of this report (the "Product"), including, but not limited to, redistribution of the Product by electronic mail, posting of the Product on a website or page, and/or providing to a third party a link to the Product, is prohibited by law and will result in prosecution. The information contained in the Product is intended solely for the recipient and may not be further distributed by the recipient to any third party.

ADDITIONAL INFORMATION IS AVAILABLE UPON REQUEST
